



University of Moratuwa

B.Sc. Engineering

Biomechanics

EMG Practical Report

Raw EMG, RMS Feature Extraction, and MVC
Normalisation

Laboratory: Biomechanics Laboratory

Assignment: Electromyography signal acquisition and interpretation

Student registration: 230449N

Data channel: EMG sensor 4, Channel 4

Sampling frequency: 4000 Hz

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1 Introduction

Electromyography (EMG) measures the electrical activity generated by skeletal muscle fibres during activation. Surface EMG is useful in biomechanics because it provides a non-invasive estimate of muscle timing, relative effort, fatigue effects, and differences between repeated movements or subjects. The recorded signal is normally bipolar and noisy, so interpretation is improved by signal processing steps such as rectification, root mean square (RMS) smoothing, and normalisation to a maximum voluntary contraction (MVC).

The objective of this practical was to acquire raw EMG signals, extract RMS features, record an MVC reference signal, and express the activity level of the movement trials relative to the MVC. The available data files are organised as raw trial recordings and an MVC sliding-RMS recording so that raw, RMS, and normalised views can be compared.

2 Methods

2.1 Apparatus

The experiment used a surface EMG sensor connected to an acquisition system, self-adhesive surface electrodes, a computer-based data capture interface, and software capable of exporting comma-separated data. The exported files indicate EMG sensor 4, Channel 4, with a sampling frequency of 4000 Hz and voltage units.

2.2 Electrode placement

The skin over the target muscle was prepared before electrode attachment to reduce contact impedance and motion artefact. The active electrodes were placed over the muscle belly and aligned with the dominant fibre direction. A reference electrode was attached to an electrically quiet bony area. The cable was secured so that movement of the wire did not introduce avoidable artefacts into the recording.

2.3 Subject instruction and safety

The subject was instructed to perform the selected motion consistently for each trial and to avoid unnecessary body movement. For MVC acquisition, the subject performed a brief maximum voluntary contraction against resistance. Rest periods were allowed between efforts to reduce fatigue. The activity was stopped if pain or discomfort occurred.

2.4 Acquisition parameters

Table 1: Data files used in the report.

File	Signal content	Nominal samples
emg_raw_trial_16.csv	Raw EMG movement trial labelled 16	82000
emg_raw_trial_19.csv	Raw EMG movement trial labelled 19	82000
emg_raw_trial_27.csv	Raw EMG movement trial labelled 27	8000
emg_raw_trial_38.csv	Raw EMG movement trial labelled 38	82000
emg_mvc_sliding_rms.csv	MVC signal after sliding RMS filtering	110000

2.5 Signal processing

The raw EMG signal $x[n]$ was inspected to identify activation bursts and artefacts. RMS processing was used because it gives a local estimate of signal power while preserving the timing envelope of muscle activation:

$$\text{RMS}[n] = \sqrt{\frac{1}{N} \sum_{k=0}^{N-1} x^2[n-k]}. \quad (1)$$

The RMS window should be long enough to smooth high-frequency oscillations but short enough to retain changes in muscle activation. In this report the interactive companion page allows the RMS window to be changed so the effect of window length can be observed directly.

Amplitude normalisation was performed using the MVC reference:

$$\%MVC[n] = \frac{\text{RMS}_{\text{trial}}[n]}{\max(\text{RMS}_{\text{MVC}})} \times 100. \quad (2)$$

This converts each trial into a relative activation level and makes comparisons less dependent on electrode contact, sensor gain, and individual signal amplitude.

3 Results

3.1 Raw EMG

The raw EMG recordings contain the original voltage-time signal from the sensor. This graph is used to inspect signal quality, baseline noise, sudden motion artefacts, clipping, and the timing of muscle activation. Raw EMG should not be used alone for comparing effort because its high-frequency oscillations make amplitude interpretation difficult.

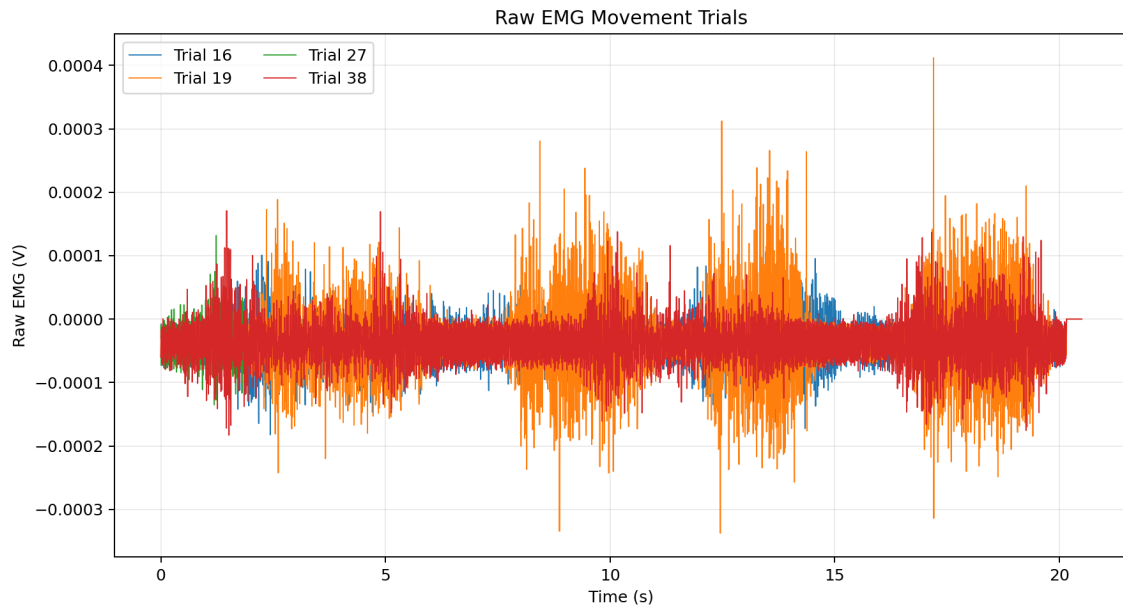


Figure 1: Raw EMG voltage traces for the four movement trials.

3.2 RMS envelope

The RMS graph is used as an activation envelope. It reduces the visual complexity of the raw waveform and provides a feature proportional to local signal energy. RMS values are suitable for comparing activation timing and relative effort across repeated movements when the same processing window is used.

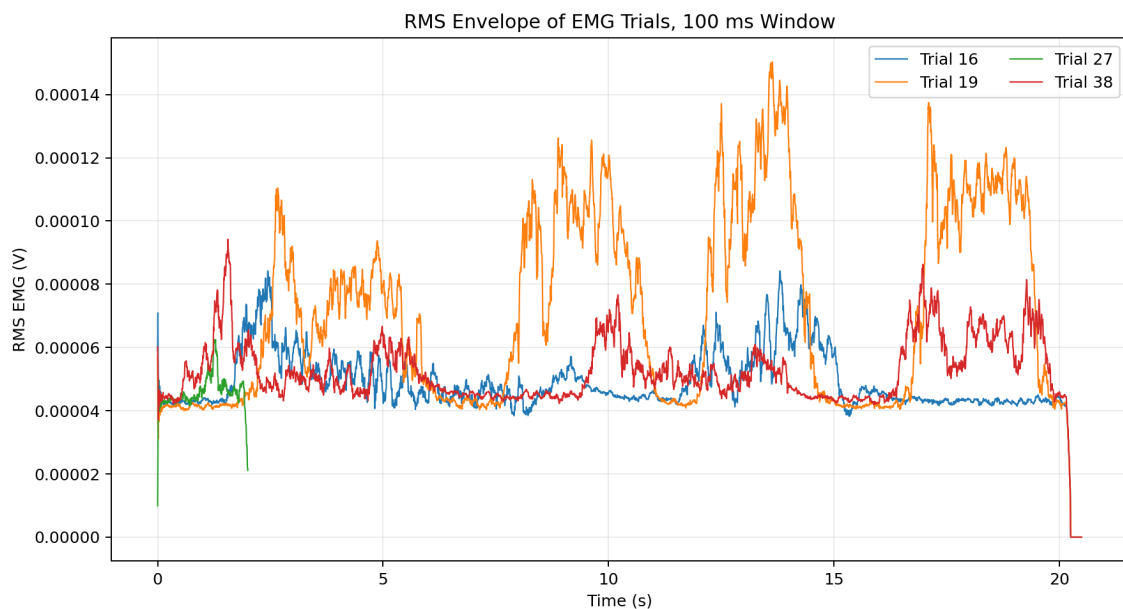


Figure 2: RMS envelopes computed from the raw EMG trials using a 100 ms moving window.

3.3 MVC-normalised amplitude

The MVC-normalised graph expresses trial activity as a percentage of the maximum voluntary contraction reference. This graph is used to compare effort between trials and subjects because it reports activity relative to a reference capacity rather than in raw volts. In the data set, `emg_mvc_sliding_rms.csv` provides the MVC reference signal.

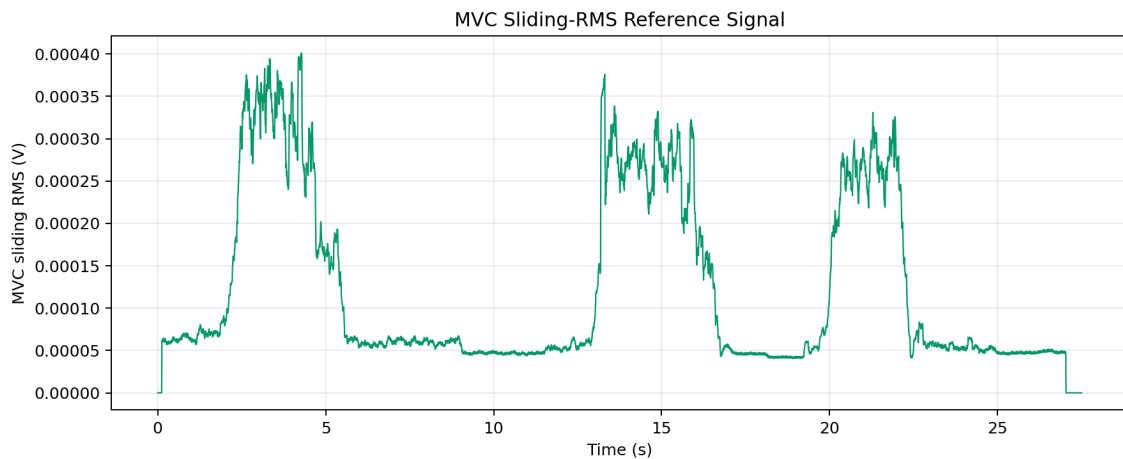


Figure 3: MVC sliding-RMS reference used to normalise the movement-trial RMS envelopes.

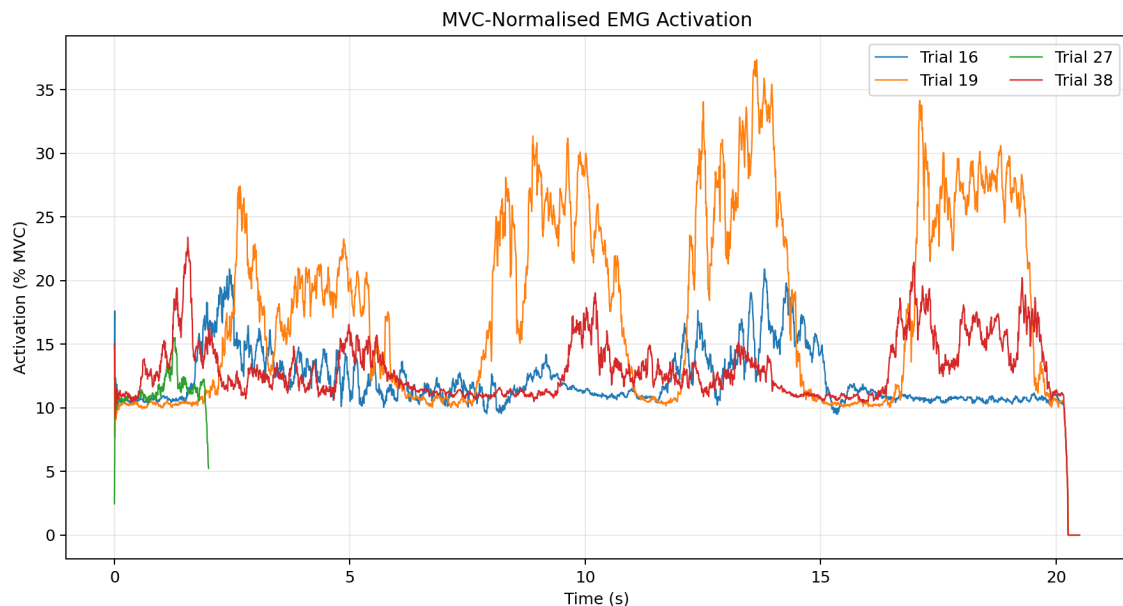


Figure 4: Movement-trial RMS envelopes normalised to the peak MVC reference and expressed as percentage MVC.

4 Discussion

The raw EMG trials show the direct sensor output and are most useful for quality control. Large abrupt spikes or baseline shifts would indicate motion artefact or poor electrode

contact. A stable baseline and repeated activation bursts indicate that the subject followed the motion instruction consistently.

RMS processing is required because EMG alternates rapidly around zero. Squaring and averaging over a moving window produces a positive envelope that represents local muscle activation. A shorter window follows rapid changes but remains noisier; a longer window produces a smoother trace but can hide short bursts. Therefore, the selected RMS window should be reported with the results.

MVC normalisation is important because raw EMG amplitude depends on electrode placement, skin impedance, adipose tissue thickness, amplifier gain, and subject-specific physiology. Dividing trial RMS by the MVC RMS reference reduces these effects and presents activation as a percentage of the subject's maximum recorded effort. This makes inter-trial and inter-subject comparisons more meaningful, provided the MVC was performed correctly and without pain.

5 Conclusion

This practical recorded surface EMG data from EMG sensor 4, Channel 4, at 4000 Hz. The report uses four raw movement trials and one MVC sliding-RMS file. Raw EMG is used to check signal quality and timing, RMS is used to estimate the activation envelope, and MVC-normalised amplitude is used to compare effort on a relative scale. The interactive HTML report provides the required raw, RMS, and normalised graphs from the organised CSV files.